

# SICKLESENSE

## Citation-Bound Clinical Evidence Assistant for Sickle Cell Disease

A grounded RAG system integrating domain-specific BiomedBERT embeddings, Azure AI Search hybrid retrieval, and constrained LLM synthesis.

Presented by Team MINOR THREAT 2026 | Creativa SCD Hackathon

# Why General-Purpose AI Fails in Sickle Cell Care

## Fragmented Guidelines

- Authoritative protocols (NHLBI, WHO, regional studies) span hundreds of pages.
- Clinicians lack fast, trustworthy margin-note tools during urgent patient rounds.
- Critical dosage and screening thresholds (e.g. TCD velocities) are buried in dense text.

## Generic LLM Hallucinations

- Standard LLMs generate ungrounded, unverified medical advice with false confidence.
- Fabricated citations and inaccurate contraindication advice create severe patient risk.
- No page-level provenance or verification checks against the true medical corpus.

## Uncontrolled Prescribing Risk

- Population guidelines must never be confused with individual patient prescriptions.
- Existing chatbots fail to intercept patient-specific inquiries (e.g. 'What dose for my child?').
- Lack of strict safety interceptors creates clinical liability.

## OUR SOLUTION

# SickleSense: The Citation-Bound Clinical Margin Note

## Strict Grounding

Every generated claim is bound to retrieved evidence chunks.  
Zero unverified external speculation.

## Verifiable Citations

Exact page numbers, section classifications, and chunk IDs (doc-CH-XXXX) for transparent provenance.

## Two-Tier Safety Gates

Regex interceptor for patient-specific dosing requests +  
score threshold gating for out-of-scope queries.

## Clinical Speed & Reliability

FastAPI Linux architecture on Azure App Service with  
sub-second hybrid retrieval and structured markdown output.

## Three Authoritative Guidelines Powering SickSense

### NHLBI (2014)

**National Heart, Lung, and Blood Institute (NIH)**

*Evidence-Based Management of SCD Report*

- Adult & pediatric hydroxyurea initiation thresholds
- Annual TCD stroke screening protocols
- Acute chest syndrome diagnosis & management
- Transfusion therapy & CKD monitoring

### JCM (2024)

**Journal of Clinical Medicine (MDPI)**

*Multicenter Retrospective Analysis (Taif Cohort)*

- Real-world Saudi multicenter cohort findings
- Splenic disease manifestations across age groups
- Hospital length of stay & ICU admission rates
- Pediatric vs. adult clinical comparison

### WHO (2026)

**World Health Organization (Geneva)**

*Consolidated Guidelines for Children & Adolescents*

- Universal newborn screening & diagnostic pathways
- Penicillin & malaria prophylaxis in endemic zones
- Universal pediatric hydroxyurea recommendation
- Growth faltering, danger signs, & referral criteria

# Parent-Child Chunking: Precision Without Losing Context

## Why Parent-Child Chunking?

- Small Chunks (800 chars): Optimize dense embedding similarity and pinpoint exact sentences in dense clinical text.
- Parent References (Page-Level): Preserve chapter context, section hierarchy, and guideline tables.
- 150-Char Overlap: Guarantees medical terms and multi-word drug protocols (e.g. '15-20 mg/kg/day') are never split across boundaries.
- Result: High Precision@3 retrieval with complete paragraph context restored during LLM synthesis.

## Indexed Vector Schema (Azure AI Search)

```
{
  "chunk_id": "nhlbi-scd-2014-CH-0396",
  "parent_id": "nhlbi-scd-2014-P0077",
  "document_id": "nhlbi-scd-2014",
  "section": "Hydroxyurea Therapy",
  "page_number": 77,
  "text": "In adults with SCA who have >= 3...",
  "vector": [0.0124, -0.0451, ..., 768 dims]
}
```

## Hybrid Search & Reciprocal Rank Fusion (RRF)

### 1. Dense Vector Embeddings

NeuML/biomedbert-base-embeddings

Transforms clinical queries into 768-dimensional normalized dense vectors. Trained on PubMed and biomedical literature to capture deep clinical semantic relationships.

### 2. Lexical BM25 Search

Exact Medical Acronym Matching

Performs inverted-index BM25 lexical search to ensure precise matching for critical medical abbreviations (VOC, ACS, TCD, HbSS, HbSβ■) that vector search might blur.

### 3. Reciprocal Rank Fusion

Server-Side Score Merging

Azure AI Search merges dense vector and BM25 rankings via RRF algorithm. Scores are calibrated against threshold (0.025) to discard irrelevant questions before generation.

# FastAPI on Azure App Service & Azure AI Search

## FastAPI Backend Architecture

- High-Performance Async API: Python service built with FastAPI and strict Pydantic schemas.
- Dedicated App Service Linux: Always-On configuration eliminates serverless cold starts.
- Groq LLM Acceleration: Queries openai/gpt-oss-120b in sub-second response times under strict temperature=0 grounding.
- Automated OpenAPI Docs: Live interactive Swagger UI at /docs and monitoring at /health.

## Azure Cloud Infrastructure

- Azure AI Search Index (creativa-hackathon-pc): Hosts 768d vector index with HNSW cosine similarity alongside BM25 indexes.
- Azure App Service (sicklesense-api): Production Linux container hosting the FastAPI backend in Sweden Central region.
- CORS & Security: Secure frontend access without exposing database credentials or API keys.
- Zero-Downtime Reliability: Fully managed scalable cloud instance.

# Vite + React SPA & GitHub Actions Deployment

## Modern React UI

### Vite + React + Lucide

- Navy + Crimson clinical branding matching SickSense mascot.
- Markdown tables with remark-gfm responsive containers.
- Live backend health checker & interactive FAQ accordion.
- Zero chat clutter: structured evidence notes.

## Decoupled Integration

### RESTful JSON Protocol

- POST /api/query: Sends { query, top\_k } payload.
- Dynamic citation mapping displaying score, section, & chunk ID.
- AbortController timeout handling (30s resilience).
- Direct sample query loading from Sources page.

## Automated CI/CD

### GitHub Actions & Pages

- Automated deployment workflow on git push to master.
- Production Vite build outputting optimized static assets.
- Relative base path configuration for reliable subpath routing.
- Zero hosting maintenance overhead.

## Empirical Evaluation Across 20 Clinical Benchmark Prompts

PRECISION@3

**0.500**

50% of top-3 retrieved passages directly answer clinical intent

PRECISION@5

**0.463**

Broad clinical context retained across 5 retrieved chunks

OUT-OF-SCOPE REJECTION

**100%**

0.000 false-relevance rate on unrelated medical questions

### Benchmark Categories Tested During Development

1. Direct Guideline Fact Lookup (e.g. 'When should hydroxyurea be started in adults?') -> Exact guideline citation matched.
2. Paraphrased Clinical Queries (e.g. 'How to lower alloimmunization risk from transfusion?') -> Correct matching across chapters.
3. Thresholds & Acronyms (e.g. 'What does TCD stand for and how is stroke screened?') -> Precise abbreviation expansion.
4. Out-of-Scope Disease Questions (e.g. 'Ibuprofen dosage for migraine in healthy adult') -> Gated & safely refused.
5. Unsafe Patient-Specific Dosage Inquiries -> Intercepted immediately before LLM invocation.

# SICKLESENSE

*"Sources Before Certainty"*

SickleSense demonstrates how domain-adapted embeddings, parent-child chunking, and hybrid RRF search can transform hundreds of pages of complex medical guidelines into an instant, traceable, and safe clinical assistant.

Live Web App: <https://O-076.github.io/sicklesense/>

FastAPI Swagger Docs: <https://sicklesense-api-dgh3aqdwb3eghdc4.swedencentral-01.azurewebsites.net/docs>

Developed with care by Team MINOR THREAT 2026 for the Creativa SCD AI Hackathon